QAOA fails if mixer includes parameterized mcrz

https://github.com/Qiskit/qiskit/issues/8301

ROW 79

QAOA fails if mixer includes parameterized mcrz #8301

New issue

Closed

ejwilson3 opened on Jul 5, 2022

Environment

- **Qiskit Terra version**: 0.21.0
- **Qiskit Optimization version**: 0.4.0
- **Python version**: 3.8.9
- **Operating system**: MacOS Big Sur 11.6.6

What is happening?

Submitting a QAOA with a custom mixer which includes a mcrz gate with a parameter (even if it's only controlled by a single qubit) it throws `CircuitError: 'Cannot bind parameters(t[1]) not present in the circuit.`

This doesn't happen with `crz` or `mcp` gates.

How can we reproduce the issue?

```
import qiskit
from qiskit.circuit import QuantumCircuit
from qiskit.algorithms import QAOA
from qiskit.algorithms.optimizers import COBYLA
from qiskit_optimization.algorithms import MinimumEigenOptimizer
quantum_instance = qiskit.utils.QuantumInstance(qiskit.Aer.get_backend('aer_simulator'))

mixer = QuantumCircuit(2)
beta = qiskit.circuit.Parameter('B')
mixer.mcrz(-1*beta, [0], 1)

# The phase separator doesn't matter, this is just an example
prog = qiskit_optimization.QuadraticProgram('example')
prog.binary_var('v1')
prog.binary_var('v2')
prog.minimize(quadratic={('v1', 'v2'): 1})

qaoa = MinimumEigenOptimizer(QAOA(mixer=mixer, optimizer=COBYLA(), reps=1, quantum_instance=quantum_instance))
result = qaoa.solve(prog)
```

Which ends up producing

```
File ~/Library/Python/3.8/lib/python/site-packages/qiskit/circuit/quantumcircuit.py:2577, in QuantumCirc
   2571 params_not_in_circuit = [
   2572     param_key
   2573     for param_key in unrolled_param_dict
   2574     if param_key not in unsorted_parameters
   2575 ]
   2576 if len(params_not_in_circuit) > 0:
-> 2577     raise CircuitError(
   2578         "Cannot bind parameters ({}) not present in the circuit.".format(
   2579             ", ".join(map(str, params_not_in_circuit))
   2580         )
   2581     )
   2583 # replace the parameters with a new Parameter ("substitute") or numeric value ("bind")
   2584 for parameter, value in unrolled_param_dict.items():

CircuitError: 'Cannot bind parameters (t[1]) not present in the circuit.'
```

What should happen?

There should be no actual output for this code; there isn't one when using `mcp` or `crz` instead of `mcrz`, and `result` should be essentially the same as it is when using `crz` instead.

Assignees
No one assigned

Labels
bug

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
No branches or pull requests

Notifications Customize

Subscribe

You're not receiving notifications from this thread.

Participants

```
    2584 for parameter, value in unrolled_param_dict.items():

CircuitError: 'Cannot bind parameters (t[1]) not present in the circuit.'
```

What should happen?

There should be no actual output for this code; there isn't one when using `mcp` or `crz` instead of `mcrz`, and `result` should be essentially the same as it is when using `crz` instead.

Any suggestions?

No response

ejwilson3 added bug on Jul 5, 2022

woodsp-ibm mentioned this on Jul 12, 2023

Parameters are not bound in the MCRY gate #9244

woodsp-ibm on Jul 12, 2023 · edited by woodsp-ibm Edits Member ···

Although this talks about QAOA I think its more to do with the QAOAAnsatz from the circuit library which has the parameters and where things would bound. Hence while at first sight I was going to put the mod algorithms label on this, to deal with it appropriately when the algorithms are moved out, I did not since I don't think its the QAOA algorithm perse.

jakelishman on Oct 20, 2023 Member ···

This should have been fixed by #11032. Feel free to reopen if that's not the case.

jakelishman closed this as completed on Oct 20, 2023

Parameters are not bound in the MCRY gate #9244

Closed

New issue

adekusar-drl opened on Dec 5, 2022

Environment

- **Qiskit Terra version**: sources(`0df0d29`)
- **Python version**: 3.8
- **Operating system**: MacOS

What is happening?

When a circuit has a parameterized `MCRY` gate and `noancilla` mode is selected by the gate, the gate parameter is not bound after calling `assign_parameters()`. This causes Aer simulator to fail when such a circuit is executed. Since the mode is chosen automatically when it is not set explicitly, the bug appears not in every circuit where `mcry` is applied. This problem may be overcome by setting `use_basis_gates` to `True`. I guess, the same issue is valid for `MCRX` and `MCRZ` gates.

How can we reproduce the issue?

Run the snippet:

```
from qiskit import QuantumCircuit
from qiskit.circuit import Parameter
from qiskit_aer import AerSimulator

qc = QuantumCircuit(3)
p = Parameter("x")
qc.mcry(p, q_controls=list(range(0, 2)), q_target=2, mode="noancilla")
qc = qc.bind_parameters({p: 5.0})
print(qc)
print(f"num parameters: {qc.num_parameters}")

simulator = AerSimulator()
simulator.run(qc)
```

What should happen?

The output:

```
                                 ┌───┐                        ┌───┐
q_0: ────────────────────────────┤ X ├─────────■──────────────┤ X ├─────────■───────────
                                 └─┬─┘         │              └─┬─┘         │
q_1: ──────────────■───────────────■───────────┼────────────────■───────────┼───────────
     ┌─────────────┴───────────┐      ┌────────┴─────────┐      ┌───────────┴────────┐
q_2: ┤ U(0.5*x,0,0,0)          ├──────┤ U(-0.5*x,0,0,0)  ├──────┤ U(0.5*x,0,0,0)     ├
     └─────────────────────────┘      └──────────────────┘      └────────────────────┘
num parameters: 0
Simulation failed and returned the following error message:
ERROR: Failed to load qobj: Unable to cast Python instance to C++ type (#define PYBIND11_DETAILED_ERROR_MESSAG
```

Any suggestions?

If I set `use_basis_gate="True"` in the snippet above everything is fine. But this is just a workaround.

adekusar-drl added bug on Dec 5, 2022

Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications Customize

Subscribe

You're not receiving notifications from this thread.

Participants

Add base-gate callback to `CUGate.params` return #11032

Merged mtreinish merged 2 commits into `Qiskit:main` from `jakelishman:workaround-cu-parameters` on Oct 19, 2023

Conversation 11 · Commits 2 · Checks 0 · Files changed 6 · +122 −32

jakelishman commented on Oct 17, 2023 • edited · Member

Summary

This causes item-setting in `CUGate.params` to forward those sets on to the `UGate` in its `base_gate` unless they are setting the `gamma` parameter, which is not shared.

`CUGate` breaks several assumptions and components of the `ControlledGate` interface, including by having more parameters than its `base_gate`; `ControlledGate` assumes that its subclasses will not have a separate `_params` field, but instead will always just view onto their `base_gate._params`. This commit changes the behaviour of the `params` getter to create a temporary list that backreferences to the base gate, satisfying the requirement.

Details and comments

This is a replacement PR for #9118, as enough of Terra had moved on since that PR that the workaround in it was no longer sufficient for most test cases. This commit builds on the work from that PR and I've credited @rrodenbusch as a co-author on it, since they solved the initial problem, and it's just my fault that the PR then staled due to lack of review.

This commit is by no means ideal, but the root problem is that `CUGate` simply does not follow the `ControlledGate` interface; it does *not* satisfy $C[U] = \left(I - |\psi\rangle\langle\psi|\right)I + |\psi\rangle\langle\psi|U$ where $|\psi\rangle$ is `ctrl_state` and U is `base_gate`. Unfortunately, the class is probably a bit too heavily used for it to be easy to make the existing `CUGate` a new `CU4Gate` or so that *doesn't* subclass `ControlledGate` (although making a special `_U4Gate` that includes a global phase in its definition to use as the `base_gate` is a possibility).

Fix #7326
Fix #7410
Fix #9627
Fix #10002
Fix #10131

Close #9118 (replaces it).

🎉 1

jakelishman added the Changelog: Bugfix label on Oct 17, 2023

jakelishman requested a review from **a team** as a code owner 2 years ago

Reviewers
mtreinish ✓

Assignees
mtreinish

Labels
Changelog: Bugfix

Projects
None yet

Milestone
0.45.0

Development
Successfully merging this pull request may close these issues.
- CU gate does not get its parameter assigned
- `quantum_info.Operator` constructor raises ...
- "Parameter" not defined in the current conte...
- Problem extracting Statevector when using a...
- CUGate's params getter does not comply wit...

Notifications · Customize
Subscribe
You're not receiving notifications from this thread.

3 participants